12TH BIOLOGY · CHAPTER 16 · FULL CHAPTER NOTES

Human Endocrine System

Topic-wise short and long questions with answers from the complete chapter

16.1 HORMONES – THE CHEMICAL MESSENGERS

Q.1 [Short] What is chemical coordination? How is it different from nervous coordination?

The nervous system carries out high-speed nervous coordination. The endocrine system controls long-term processes like growth, metabolism and reproduction by sending chemical instructions, in the form of hormones, to the target body parts. This is called chemical coordination.

Q.2 [Short] What is a hormone? Why are hormones called chemical messengers?

A hormone is a regulatory chemical that is secreted into the blood or tissue fluid by an endocrine gland. The blood carries the hormone to every cell of the body, but it shows its effects only in its specific **target cells**. When the target cells receive a hormone, they respond in a specific way. That is why hormones are called **chemical messengers** or signalling molecules.

Q.3 [Short] Differentiate between exocrine and endocrine glands.

Glands are tissues that produce and release secretions. **Exocrine glands** release their secretions into ducts attached with them. **Endocrine glands** are ductless; they release their secretions, called hormones, directly into blood or tissue fluid.

Q.4 [Short] Who discovered hormones, and how?

In 1902, two British scientists, Bayliss and Starling, discovered hormones during their research on the digestive system. They found that when acidic gastric juice comes into the small intestine, a substance (which they called “secretin”) is released into the blood. Secretin reaches the pancreas and stimulates it to secrete pancreatic juice.

Q.5 [Long] Describe the chemical nature of hormones with examples.

Hormones are grouped into the following chemical categories.

1. Peptide and protein hormones

These are made of chains of amino acids. They are water-soluble, so they dissolve easily in blood. However, they cannot pass through cell membranes. So they bind to receptors on the cell surface.

Hormone	Source (gland)	Main function
Insulin	Pancreas	Lowers blood glucose level
Glucagon	Pancreas	Raises blood glucose level
Growth hormone	Anterior pituitary	Stimulates growth and cell reproduction

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Hormone	Source (gland)	Main function
Oxytocin	Hypothalamus (released by posterior pituitary)	Stimulates muscle contractions during childbirth

2. Glycoprotein hormones

Glycoproteins are conjugated proteins. They consist of a protein combined with a carbohydrate (sugar) group. They are large, complex and polar molecules, so they also cannot pass through the cell membrane.

Hormone	Source (gland)	Main function
Thyroid stimulating hormone	Anterior pituitary	Stimulates the thyroid gland to produce thyroxine
Follicle stimulating hormone	Anterior pituitary	Stimulates sperm production or egg development
Luteinizing hormone	Anterior pituitary	Triggers ovulation in females and secretion of testosterone in males

3. Amino acid derivative hormones

These are small molecules derived from a single amino acid (usually tyrosine or tryptophan). Their ability to pass through cell membranes depends on whether they behave more like a protein or more like a lipid. Epinephrine and norepinephrine cannot cross membranes, while thyroxine and tri-iodothyronine can cross.

Hormone	Source (gland)	Main function
Epinephrine (adrenaline)	Adrenal medulla	Immediate “fight or flight” response
Norepinephrine (noradrenaline)	Adrenal medulla	Increases blood pressure and alertness
Thyroxine (T_4)	Thyroid gland	Controls the speed of metabolism
Triiodothyronine (T_3)	Thyroid gland	Regulates body temperature and heart rate

4. Steroid hormones

These hormones are derived from cholesterol (a type of lipid). Unlike proteins, steroids are lipid-soluble (fat-soluble). So they can move through the cell membrane and give instructions directly to the nucleus.

Hormone	Source (gland)	Main function
Cortisol	Adrenal cortex	Regulates metabolism and helps the body respond to stress
Aldosterone	Adrenal cortex	Helps the kidneys manage salt and water balance

Hormone	Source (gland)	Main function
Testosterone	Testes	Regulates male reproductive tissues and traits
Estrogen / Progesterone	Ovaries	Regulate the female reproductive system and cycle

Q.6 [Short] Why do protein hormones bind to receptors on the cell surface while steroid hormones enter the cell?

Protein hormones are water-soluble and cannot pass through the lipid cell membrane, so they bind to receptors on the cell surface. Steroid hormones are derived from cholesterol and are lipid-soluble, so they move through the cell membrane and give instructions directly to the nucleus.

Q.7 [Short] Which amino acid derivative hormones can cross the cell membrane and which cannot?

Epinephrine and norepinephrine cannot cross cell membranes. Thyroxine and tri-iodothyronine can cross them.

Q.8 [Short] Describe the path of the chemical message carried by hormones.

Endocrine glands secrete their hormones on receiving particular stimuli. They release hormones into the blood, which carries them to all cells. Each hormone affects only its target cells, because target cells have **receptors** and each receptor binds to a specific hormone. This binding triggers events that lead to changes within the cell. Receptors can be found on the cell membrane, in the cytoplasm or in the nucleus of a cell.

Q.9 [Short] Compare hormones and neurotransmitters.

Both are chemical messengers. However, hormones are part of the endocrine system, while neurotransmitters are part of the nervous system. Hormones are often slower acting and have longer effects than neurotransmitters.

16.2 ENDOCRINE SYSTEM OF MAN

Q.10 [Short] What does the human endocrine system consist of? Name its major glands.

It consists of glands and organs that produce and secrete hormones for regulating various body functions. The major glands are the hypothalamus, pituitary, thyroid, parathyroids, pancreas (islets of Langerhans), adrenal glands and gonads (ovaries and testes). The thymus and pineal glands are also endocrine glands.

Q.11 [Long] Explain the neurosecretory role of the hypothalamus.

Many endocrine glands of the body are controlled by the master gland, i.e., the pituitary gland. However, the master gland is itself controlled by a part of the brain, i.e., the **hypothalamus**. The hypothalamus contains special **neurosecretory cells**, which not only conduct nerve impulses but also secrete hormones.

Control of posterior pituitary

Two hormones, oxytocin and antidiuretic hormone (ADH), are made by the neurosecretory cells of the hypothalamus. The axons of these cells transport oxytocin and ADH to the posterior pituitary, where they are stored for eventual release into the blood.

Control of anterior pituitary

Blood vessels connect the hypothalamus with the anterior pituitary. Neurosecretory cells of the hypothalamus secrete hormones that travel to the anterior pituitary through these blood vessels. These hormones are of two kinds:

- **Releasing hormones** stimulate the anterior pituitary to make and secrete hormones.
- **Inhibiting hormones** inhibit the production and secretion of anterior pituitary hormones.

Hormone of hypothalamus	Effect on anterior pituitary
Growth hormone releasing hormone	Secretion of growth hormone
Somatostatin	Inhibition of growth hormone
Thyrotropin releasing hormone (TRH)	Secretion of thyroid stimulating hormone (TSH)
Adreno-corticotropin releasing hormone	Secretion of adrenocorticotrophic hormone
Prolactin inhibiting factor (actually dopamine)	Inhibits secretion of prolactin
Gonadotropin releasing hormone	Secretion of FSH and LH

Q.12 [Short] What are neurosecretory cells?

They are special cells of the hypothalamus which not only conduct nerve impulses but also secrete hormones.

Q.13 [Short] Differentiate between releasing hormones and inhibiting hormones.

Both are secreted by the hypothalamus. Releasing hormones stimulate the anterior pituitary to make and secrete its hormones, e.g., TRH causes the secretion of TSH. Inhibiting hormones inhibit the production and secretion of anterior pituitary hormones, e.g., somatostatin inhibits growth hormone.

Q.14 [Short] Describe the structure of the pituitary gland. Why is it called the master gland?

The pituitary gland is attached to the hypothalamus by a stalk called the **infundibulum**. It is divided into three lobes: the anterior pituitary, the posterior pituitary and the intermediate (median) pituitary. It is called the master gland because many of its hormones regulate the activity of other endocrine glands.

Q.15 [Long] Describe the hormones of the anterior pituitary and their functions.

1. **Growth hormone (GH, or somatotropin):** stimulates the growth of muscles, bones (indirectly) and other tissues. It is also essential for proper metabolic regulation.
2. **Adrenocorticotropic hormone (ACTH, or corticotropin):** stimulates the adrenal cortex to secrete corticosteroids (cortisone and aldosterone), which regulate glucose and salt homeostasis.
3. **Thyroid-stimulating hormone (TSH, or thyrotropin):** stimulates the thyroid gland to produce thyroxine, which in turn stimulates oxidative respiration.
4. **Luteinizing hormone (LH):** in females, it stimulates the ovaries to produce progesterone and estrogen and also initiates ovulation. In males, it stimulates the testes to produce testosterone.
5. **Follicle-stimulating hormone (FSH):** stimulates the development of ovarian follicles in females and the development of sperms in males. FSH and LH are both called gonadotropins.
6. **Prolactin:** stimulates the mammary glands to produce milk. It is produced in both males and females but functions only in females.

Q.16 [Short] What are gonadotropins?

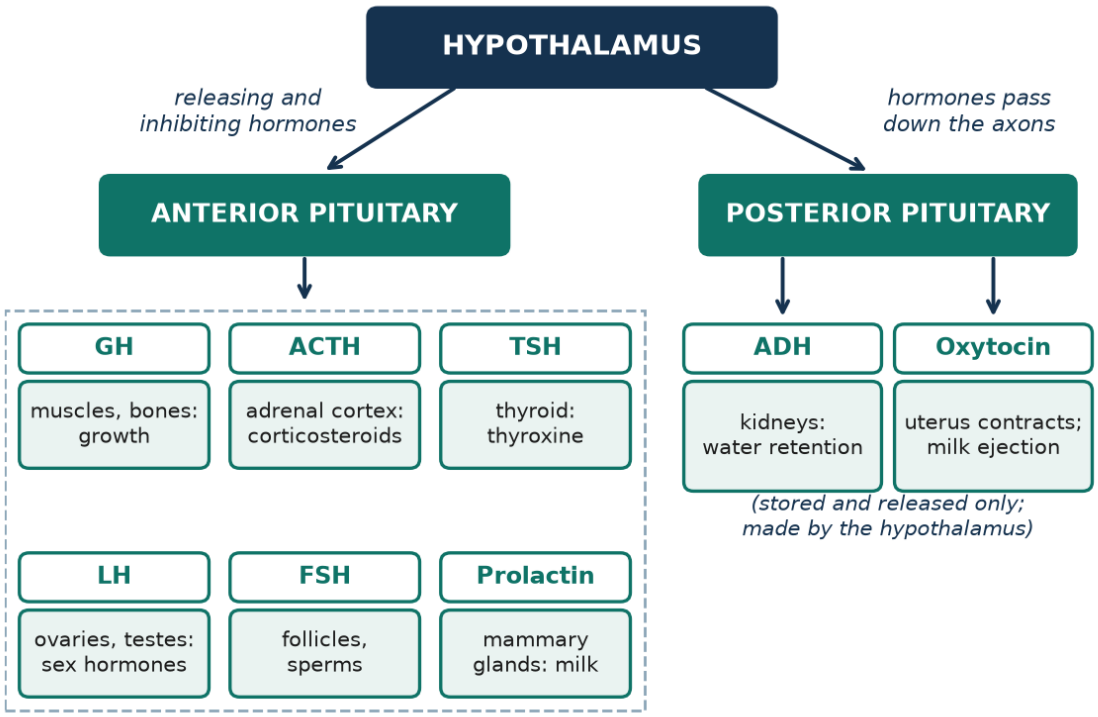
FSH and LH are both called gonadotropins, because they act on the gonads (ovaries and testes).

Q.17 [Short] Write a note on the posterior pituitary.

The posterior pituitary contains axons that originate in the hypothalamus. It does not synthesize any hormone; it only stores and releases the hormones made by the hypothalamus:

1. **Antidiuretic hormone (ADH or vasopressin):** stimulates water retention by the kidneys.
2. **Oxytocin:** stimulates milk ejection by contracting the smooth muscles of the mammary glands. It also stimulates uterine contractions in women during childbirth.

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Hormones of the pituitary gland and their targets

Q.18 [Short] Why do the hypothalamus and the posterior pituitary remain connected by a tract of axons?

During embryonic development, the posterior pituitary is formed from an outgrowth of neural tissue. That is why the hypothalamus of the brain and the posterior pituitary remain interconnected by a tract of axons.

Q.19 [Short] Why does the uterus of a breast-feeding mother return to its normal size more quickly?

Oxytocin secretion continues after childbirth in breast-feeding mothers. Oxytocin stimulates uterine contractions. So the uterus of a nursing mother returns to its normal size more quickly than that of mothers who do not breast-feed.

Q.20 [Short] What is the function of the intermediate pituitary?

In humans, the intermediate pituitary is a thin layer of cells between the anterior and posterior pituitary. It produces **melanocyte stimulating hormone (MSH)**, which stimulates the production and release of melanin by melanocytes in skin and hair.

Q.21 [Long] Describe the abnormalities caused by the imbalance of pituitary hormones.

1. **Gigantism:** over-secretion of GH in a growing child. The height is well above average, with enlarged hands, feet and facial features.
2. **Acromegaly:** over-secretion of GH in adult life. Bones can no longer increase in length but grow in thickness. Hands, feet, skull, nose and jawbone become enlarged.

3. **Dwarfism:** deficiency of GH. Development is much slower and the individual has short stature. However, the body parts stay in proportion, and brain development and IQ are unaffected.
4. **TSH:** over-secretion causes hyperthyroidism (excess of thyroxine) and under-secretion causes hypothyroidism (lack of thyroxine).
5. **ADH:** under-secretion causes diabetes insipidus (excessive production of dilute urine and frequent thirst). Over-secretion may lead to kidney problems.
6. **Oxytocin:** over-secretion causes rupturing of the uterine wall, while under-secretion inhibits the normal labour process.

Q.22 [Short] Differentiate between gigantism and acromegaly.

Both are caused by over-secretion of growth hormone. Gigantism occurs in a growing child: the height is well above average. Acromegaly occurs in adult life: bones cannot grow in length but grow in thickness, so hands, feet, skull, nose and jawbone become enlarged.

Q.23 [Short] Is growth hormone treatment suitable for every short person?

No. GH treatment is given to children who have short stature due to GH deficiency, and it improves their growth rate. But every dwarf person does not have GH deficiency, so GH therapy may not be appropriate for everyone. Doctors also do not support the use of GH for building muscle or slowing aging.

Q.24 [Long] Describe the location, structure and hormones of the thyroid gland.

The thyroid is a butterfly-shaped gland located on either side of the trachea in the front of the neck. It is composed of two connected lobes. It secretes the following hormones:

1. **Tetra-iodothyronine (T_4) or thyroxine and tri-iodothyronine (T_3):** The thyroid produces much more T_4, which is eventually converted into T_3, the more active and powerful form. These hormones increase heart rate and breathing rate. They stimulate cellular respiration and set the body's basal metabolic rate. They also enhance glucose catabolism and the synthesis of cholesterol in the liver. In children, they promote the growth of the skeleton and the development of the brain.
2. **Calcitonin:** It maintains the proper level of calcium in blood. When the blood calcium concentration rises too high, calcitonin stimulates the absorption of calcium into bones, thus lowering its level in blood.

Q.25 [Short] Why is the thyroid gland called the "Master Controller of Metabolism"?

Because its hormones dictate how fast or slow every cell in the body works.

Q.26 [Long] What are the major diseases due to the imbalance of thyroid hormones?

1. **Hypothyroidism** (deficiency of thyroid hormones): causes growth retardation, lethargy, weight gain, constipation, high cholesterol, and low heart rate and body temperature.

2. **Cretinism:** hypothyroidism during foetal and childhood development; a condition of stunted growth and mental retardation.
3. **Myxoedema:** a rare, life-threatening complication of severe hypothyroidism. Mucopolysaccharides build up in the connective tissues, causing swelling and thickening of the skin and tissues.
4. **Goitre:** swelling of the thyroid gland; it results when hypothyroidism is caused by iodine deficiency.
5. **Hyperthyroidism** (excessive secretion of thyroid hormones): causes weight loss, and high blood pressure, heart rate and body temperature.

Q.27 [Short] What is Graves' disease?

It is an autoimmune disease that causes hyperthyroidism. It often results in an enlarged thyroid. Its symptoms include irritability, muscle weakness, sleeping problems, fast heartbeat, poor tolerance of heat, diarrhoea, weight loss and thickening of skin on the shins. Exophthalmia (protrusion of the eyeballs) may also result.

Q.28 [Short] Write a note on parathyroid glands.

These are four small glands attached to the posterior surface of the thyroid gland. They secrete the hormone **parathormone**. It is released when the blood calcium level falls. Parathormone:

- stimulates the osteoclasts (bone cells) to dissolve the calcium crystals of the bone matrix and release calcium into the blood;
- stimulates the kidneys to reabsorb calcium from the urine;
- causes the activation of vitamin D, which is needed for the absorption of calcium from food in the intestine.

Q.29 [Short] What happens due to the imbalance of parathormone?

Over-secretion of parathormone (which may be due to a parathyroid tumour) causes excessive release of calcium from bones. It leads to deformed and soft bones. The blood calcium level rises (hypercalcemia), which depresses the nervous system and causes weakness of muscles.

Under-secretion causes hypocalcaemia. This increases the excitability of neurons. It can also lead to **tetany**, in which muscles remain in a contracted state.

Q.30 [Short] How do calcitonin and parathormone together control the blood calcium level?

They are antagonistic. When blood calcium rises too high, calcitonin from the thyroid stimulates the absorption of calcium into bones and lowers its level. When blood calcium falls, parathormone from the parathyroids releases calcium from bones, makes the kidneys reabsorb calcium and activates vitamin D, so the level rises.

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Q.31 [Long] Write a note on the pancreas as an endocrine gland.

The pancreas mostly contains exocrine cells, which secrete pancreatic juice into the pancreatic duct. But it also has specialized cells which function as an endocrine gland. These cells are distributed as patches called **islets of Langerhans**. Each islet has two types of cells: **beta (β) cells**, which are larger in number and secrete insulin, and **alpha (α) cells**, which secrete glucagon.

Insulin reduces the blood glucose level. It is secreted when the level of blood glucose rises. It increases the rate of glucose uptake by body cells (e.g., skeletal muscles and fat cells). It promotes glycogenesis (conversion of glucose to glycogen) in the liver, increases the use of glucose in cellular respiration, promotes the conversion of excess glucose to fats and inhibits gluconeogenesis (glucose synthesis).

Glucagon raises the blood glucose level. It is secreted when the blood glucose level is below normal. It mainly promotes the hydrolysis of stored glycogen in the liver. As a result, glucose is released into the blood and can be taken up by cells and used for energy.

Q.32 [Short] What are the effects of the imbalance of pancreatic hormones?

1. **Diabetes mellitus (hyperglycaemia):** caused by under-secretion of insulin. Cells are unable to obtain glucose from blood, so the blood glucose level is above normal. The excess glucose inhibits water reabsorption by the kidneys, so large amounts of urine are produced. It leads to dehydration and kidney damage.
2. **Hypoglycaemia:** caused by over-secretion of insulin. The blood glucose level falls below normal. Symptoms include lethargy, dizziness, nervousness and, in extreme cases, unconsciousness.

Q.33 [Short] Differentiate between Type I and Type II diabetes mellitus.

Type I diabetes	Type II diabetes
Insulin-dependent diabetes mellitus.	Non-insulin dependent diabetes mellitus.
The patients lack insulin-secreting β cells.	The patients have normal insulin levels, but their cells have a reduced sensitivity to insulin.
Treated with daily injections of insulin.	Controlled through other medicines, diet and exercise; insulin is not required.

Q.34 [Short] Why is insulin injected and not taken orally?

Insulin is a peptide hormone. If it is taken orally, it would be digested in the digestive tract. So it must be injected into the blood. Injectable insulin is produced by genetically engineered bacteria.

Q.35 [Long] Write a detailed note on the adrenal glands and their hormones.

One adrenal gland is located just above each kidney. Each gland is composed of an inner portion, the **adrenal medulla**, and an outer layer, the **adrenal cortex**.

Adrenal medulla

It produces two hormones: **epinephrine** (adrenaline) and **norepinephrine** (noradrenaline). When a person is stressed, the medulla secretes these hormones. They prepare the body for "fight or flight". In this response there is increased heart rate, increased blood pressure, dilation of the bronchioles, elevation in blood glucose, and reduced blood flow to the skin and digestive organs.

Adrenal cortex

The hormones of the adrenal cortex are called **corticosteroids**. These include:

1. **Glucocorticoids**, e.g., cortisol and related steroids. They maintain glucose homeostasis. They stimulate the breakdown of muscle protein into amino acids, which are carried by blood to the liver. They also stimulate the liver for gluconeogenesis (conversion of amino acids into glucose).
2. **Mineralocorticoids**, e.g., aldosterone. It helps to regulate mineral balance in blood. It stimulates the kidneys to reabsorb sodium from urine; in this way water is also reabsorbed, so normal blood volume and pressure are maintained. It also stimulates the kidneys to secrete the excess of potassium into urine.

Q.36 [Short] How is the secretion of the adrenal medulla and the adrenal cortex controlled?

Secretion of hormones from the adrenal medulla is controlled by the nervous system. Secretion of hormones from the adrenal cortex is controlled by the hormones of the anterior pituitary (ACTH).

Q.37 [Short] Describe the disorders caused by the imbalance of adrenal hormones.

1. **Adrenal medulla:** over-secretion may cause hypertension and aggressive behaviour during routine life. Under-secretion causes failure to combat an emergency situation.
2. **Cushing disease:** over-secretion of cortisol. It occurs when the pituitary gland produces too much ACTH. There is excessive protein breakdown. It results in high blood sugar, high blood pressure, thin bones, muscle loss, thin and fragile skin, depression and too much facial hair (in women).
3. **Addison's disease:** under-secretion of cortisol. It is characterized by general metabolic disturbance, weakness of muscles and loss of salts. The symptoms usually appear slowly.

Q.38 [Short] What are gonads? Which pituitary hormones control them?

Gonads (ovaries in females and testes in males) are gamete-producing organs. They also produce sex hormones under the influence of LH and FSH of the anterior pituitary.

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Q.39 [Long] Describe the hormones of the ovaries and their functions.

1. **Estrogen:** secreted by the Graafian follicle in response to FSH. Its functions include the development of secondary sexual characteristics (e.g., breast development, widening of the pelvis, fat distribution in hips and thighs), regulation of the menstrual cycle, repair of the uterine lining after menstruation, ovulation, maintenance of bone density, lowering of “bad” cholesterol (LDL), maintenance of healthy blood pressure, and regulation of mood and memory.
2. **Progesterone:** produced by the corpus luteum in response to LH during the menstrual cycle, and by the placenta during pregnancy. It inhibits further FSH secretion from the pituitary, so no more follicles ripen. It causes further thickening and vascularisation of the uterus wall for maintaining pregnancy. It suppresses ovulation.

Q.40 [Short] Write a note on testosterone.

LH stimulates the testes to secrete a group of sex hormones called androgens. **Testosterone** is an androgen secreted by the interstitial cells among the seminiferous tubules. It initiates the maturation of male reproductive organs and the appearance of secondary sex characteristics and sex drive. It is also necessary for normal sperm production.

Q.41 [Short] What is puberty?

Puberty is the adolescent stage during which the sex organs mature and secondary sex characteristics appear. In males, sperm production begins, the voice deepens, the chest broadens and hair grows on the body and face. In females, the menstrual cycle begins, the breasts grow and the hips widen.

Q.42 [Short] What are the effects of the imbalance of the hormones of gonads?

1. **Estrogen:** deficiency in young females means they fail to mature sexually; deficiency in adults leads to sterility. Over-secretion may lead to fibroids (abnormal growth) in the uterus and polycystic ovaries.
2. **Progesterone:** under-secretion during the menstrual cycle decreases the chance of pregnancy and may cause early menstruation. Under-secretion during pregnancy may lead to miscarriage.
3. **Testosterone:** under-secretion in males causes the development of feminine characteristics and male sterility.

Q.43 [Short] What are anabolic steroids? Why is their use in sports banned?

Artificially synthesized steroids, also known as **anabolic steroids**, are synthetic versions of testosterone. Athletes have widely used them to enhance performance in sports, because they increase muscle mass, strength and endurance. However, their use in sports is illegal and banned. Their long-term effects are:

- Damage to the liver, leading to hepatitis and liver tumours.
- Increased risk of heart disease, stroke and blood clots.

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- Disruption of the body's natural hormone balance, leading to infertility, impotence and breast development in men.
- Mental health issues such as depression, anxiety and aggressive behaviour.

Q.44 [Long] Describe the hormones of the thymus, pineal gland, digestive organs and kidneys.

Gland / tissue	Hormone	Function
Thymus gland (beneath the sternum, between the lungs)	Thymosin	Stimulates the maturation of T cells, which help defend the body from pathogens
Pineal gland (near the base of brain)	Melatonin	Regulates sleep patterns
Stomach wall	Gastrin	Stimulates the gastric glands to release gastric juice
Small intestine wall	Secretin	Stimulates the pancreas to release pancreatic juice
Small intestine wall	Cholecystokinin	Stimulates the gallbladder to release bile
Kidneys	Renin	Controls the production of aldosterone in adrenal glands
Kidneys	Erythropoietin	Stimulates bone marrow for the production of RBCs
Kidneys	Calcitriol	Increases blood calcium level by absorption of calcium from intestines

Q.45 [Short] What are endorphins and prostaglandins?

Endorphins are hormones produced by the brain. They bind to pain receptors and so block the sensation of pain. **Prostaglandins** are hormone-like substances produced by almost all cells of the body. They are powerful, locally-acting vasodilators and are involved in inflammation.

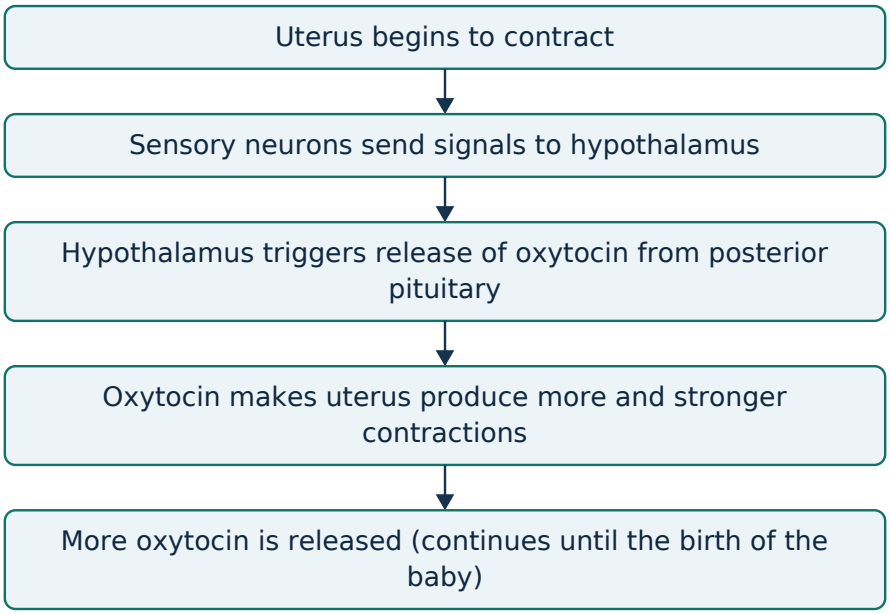
16.3 FEEDBACK MECHANISM IN ENDOCRINE SYSTEM

Q.46 [Short] What is a feedback mechanism?

It is the phenomenon in which a controlling mechanism is itself controlled by the product of the process it is controlling. There are two types: positive feedback and negative feedback.

Q.47 [Long] Explain positive feedback with reference to oxytocin.

In positive feedback, the product or response made by the controlling system **speeds up** the action of the controlling system. An example is the release of oxytocin by the posterior pituitary during childbirth.



Positive feedback of oxytocin

Q.48 [Long] Explain negative feedback with reference to insulin and glucagon.

In negative feedback, the end product **stops** the initial step of the process. It is common in the endocrine system.

Insulin: When blood glucose goes above normal, the beta cells of the islets of Langerhans sense it and release insulin. Insulin lowers the blood glucose level to normal. This result, i.e., the normal blood glucose level, makes the beta cells stop releasing insulin.

Glucagon: When blood glucose goes low, the alpha cells of the islets of Langerhans secrete glucagon. Glucagon raises the blood glucose level to normal. This result, i.e., the normal blood glucose level, makes the alpha cells stop releasing glucagon.

Q.49 [Short] Why is negative feedback more suitable than positive feedback for keeping blood glucose normal?

In negative feedback, the result (normal blood glucose) stops the release of the hormone, so the level returns to normal and stays there. In positive feedback, the response would speed up the hormone release more and more, and the level would move further away from normal.